

Living Technical Notes on Grid Intelligence

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1. Living Technical Notes on Grid Intelligence

These are living graduate-level technical notes on mathematics, control systems, machine learning, power systems, optimization, PyTorch research implementation, and parallel scientific computing.

The notes are developed as part of a long-term study and research program focused on **grid intelligence**, including power-flow modeling, optimal power flow, learning-based control, scientific machine learning, and scalable simulation workflows.

1.1. Scope

The notes are organized into six major parts:

Structure

1. **Mathematical Foundations**

Proofs, analysis, probability, optimization, dynamical systems, and functional tools.

2. **Control Systems**

Linear control, nonlinear control, optimal control, MPC, robustness, and learning-based control.

3. **Machine Learning, Reinforcement Learning, and AI**

ML foundations, probabilistic learning, deep learning, RL, scientific ML, and modern LLM systems.

4. **Power Systems, Power Flow, and OPF**

AC power flow, distribution modeling, OPF, convex relaxations, DERs, batteries, and hosting capacity.

5. **PyTorch for Research Implementation**

Autograd, neural networks, sequence models, transformers, generative models, RL, and scientific ML.

6. **Parallel Computing and Scientific Software Performance**

CPU optimization, multiprocessing, CUDA, benchmarking, and accelerated grid analytics.

1.2. Status

These notes are not a polished textbook. They are evolving technical notes generated from structured study sessions. Over time, selected sections may be reorganized into polished tutorials, dissertation-support material, internal reports, or book-length manuscripts.

1.3. Intended Use

These notes are meant to support:

- rigorous self-study,
- PhD research development,
- iGridSim-related technical work,
- implementation of ML-assisted OPF and grid analytics,
- long-term preparation for research in learning, control, optimization, and scientific computing.

2. Preface

These notes are meant to support long-term graduate-level study and research development in grid intelligence, optimization, machine learning, control, and scientific computing.

The current version should be viewed as a living technical notebook rather than a finished book. Over time, selected material may be reorganized into polished tutorials, dissertation-support material, internal technical reports, or book-length manuscripts.

3. How to Read These Notes

The notes are organized by major study tracks rather than by final textbook structure.

Each chapter may include:

- motivation
- definitions
- mathematical derivations
- algorithms
- worked examples
- implementation notes
- exercises
- references

The progress dashboard is included for tracking study and notes-writing progress. Technical chapters should be read as evolving notes.

4. Progress Dashboard

This page tracks study progress and notes-writing progress across the main study tracks.

💡 How to interpret this table

- **Study percent** tracks conceptual/course progress.
- **Notes percent** tracks how much has been written into these technical notes.
- The dashboard is mainly for internal tracking.

Table 4.1.

	part	project	chapter	status	study_percent	notes_percent
0	Part I	Mathematical Foundations	Proof Writing and Real Analysis	Not started	0%	0%
1	Part II	Control Systems	Control Modeling	Not started	0%	0%
2	Part III	ML/RL/AI Foundations	ML Foundations	Not started	0%	0%
3	Part IV	Power Systems and OPF	Phasors Per-Unit and Complex Power	Not started	0%	0%
4	Part V	PyTorch	Computational Graphs and Autodiff	Not started	0%	0%
5	Part VI	Parallel Computing	Performance Foundations	Not started	0%	0%

5. Notation

This page collects common notation used throughout the notes.

Symbol	Meaning
x	State vector
u	Control input
y	Output vector
A, B, C, D	State-space matrices
P	Real power
Q	Reactive power
S	Complex power
V	Voltage phasor
I	Current phasor

Part I.

Part I — Mathematical Foundations

6. Chapter 1 — Proof Writing and Real Analysis

6.1. Motivation

6.2. Logic, Statements, and Quantifiers

6.3. Direct Proofs

6.4. Proof by Contradiction and Contrapositive

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6.6. Continuity and Compactness

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Part II.

Part II — Control Systems

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7.1. Motivation

7.2. Continuous-Time and Discrete-Time Dynamics

7.3. States, Inputs, Outputs, and Disturbances

7.4. Equilibrium Points

7.5. Linearization Around Equilibria

7.6. Control Objectives

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Part III.

**Part III — Machine Learning, Reinforcement
Learning, and AI**

8. Chapter 1 — Mathematical Foundations for ML/RL/AI

8.1. Motivation

8.2. Linear Algebra for Learning

8.3. Matrix Calculus

8.4. Probability and Statistics

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8.6. Information-Theoretic Quantities

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Part IV.

**Part IV — Power Systems, Power Flow, and
OPF**

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9.1. Motivation

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9.7. OpenDSS and OPF Connections

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Part V.

**Part V — PyTorch for Research
Implementation**

10. Chapter 1 — Computational Graphs and Automatic Differentiation

10.1. Motivation

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Part VI.

**Part VI — Parallel Computing and Scientific
Software Performance**

11. Chapter 1 — Performance Foundations

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11.5. Brent's Theorem

11.6. Memory Cost and Roofline Intuition

11.7. Benchmarking Methodology

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11.9. References

12. Software Setup

This appendix records the software setup used for building and maintaining these notes.

12.1. Quarto

12.2. Python

12.3. Git and GitHub

12.4. Optional LaTeX / PDF Toolchain

13. References